

PE-EC703C — Wavelet Transforms

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Module 5 Exam Notes: Wavelet Applications & Beyond Wavelets

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1. Transform Coding and Data Compression

Data compression reduces transmission bandwidth and storage requirements. In **Transform Coding**, a mathematical transform maps spatial/temporal domain signal values to a frequency-like domain, where the coefficients are less correlated and easier to compress.

1.1 Principles of Transform Coding

Transform coding relies on three core steps:

1. **Transform:** De-correlates the signal data, packing most of the energy into a few high-magnitude coefficients (Energy Compaction).
2. **Quantization:** Converts continuous or high-precision coefficients to discrete levels. This is a lossy step where small, high-frequency coefficients (representing details less perceptible to human eyes/ears) are rounded to zero.
3. **Entropy Coding:** Applies lossless compression (e.g., Huffman or Arithmetic coding) to the quantized symbols, mapping more frequent values to shorter codewords.

1.2 Image and Audio Compression: DWT/DTWT vs. DCT (JPEG 2000 vs. JPEG)

For discrete-time sequences (such as digital audio and images), the Discrete Wavelet Transform is implemented as the **Discrete-Time Wavelet Transform (DTWT)** using digital filter banks.

- **JPEG (Discrete Cosine Transform):** Splits an image into non-overlapping 8×8 blocks and applies 2D DCT to each block. At high compression ratios, discarding high-frequency coefficients creates artificial grid-like discontinuities at block boundaries, known as **blocking artifacts**.
- **JPEG 2000 (Discrete Wavelet Transform):** Applies DWT globally over the entire image (or large tile regions). This eliminates blocking artifacts, resulting in graceful, blurry degradation rather than grid patterns.
- **Key Advantages of JPEG 2000 over JPEG:**
 - **No Blocking Artifacts:** Smooth, visually superior reconstruction at low bitrates.
 - **Progressive Transmission:** Single bitstream contains multiple resolutions and quality layers. A low-resolution thumbnail is decoded first, followed by details as more data arrives.
 - **Region of Interest (ROI) Coding:** Particular regions of an image (e.g., a tumor in a medical scan) can be encoded with higher precision than the background.
 - **Unified Lossless and Lossy:** Uses the CDF 5/3 filter bank for lossless compression and the CDF 9/7 filter bank for lossy compression.

Feature	Standard JPEG (DCT-based)	JPEG 2000 (DWT-based)
Core Transform	Block-based Discrete Cosine Transform (DCT).	Discrete Wavelet Transform (DWT).
Processing Unit	Fixed 8×8 pixel blocks.	Entire image or large tile regions.
Compression Ratio	Poor quality at ratios $> 30 : 1$ (blocks visible).	Excellent quality at high ratios ($> 100 : 1$).
Artifact Type	Blocking artifacts (grid patterns) and mosquito noise.	Blurring and ringing artifacts around edges.
Scalability	Not natively supported (needs separate files).	Resolution and quality scalability in one stream.
Entropy Coder	Huffman Coding.	Context-adaptive EBCOT (Arithmetic Coding).

1.3 Audio Coding and Video Coding

- **Audio Coding:** The human ear acts as a spectrum analyzer. Wavelet-based audio coders exploit this multi-resolution structure. By dividing the audio spectrum into subbands matching the ear's **critical bands**, subband coding allocates bits dynamically based on psychoacoustic masking (louder sounds mask quieter adjacent frequencies).
- **Video Coding:** Standard video codecs (H.264/H.265) utilize block-based motion estimation and DCT. Wavelet-based video coding uses DWT on individual frames or applies a **3D DWT** (spatial x - y DWT + temporal t DWT) to compress spatial and motion information simultaneously, avoiding motion compensation loops.

2. Wavelet De-noising and Image Processing

2.1 Wavelet De-noising (Hard vs. Soft Thresholding)

Consider a noisy signal model:

$$y_i = x_i + \sigma z_i, \quad i = 1, 2, \dots, N$$

where y_i is the observed noisy signal, x_i is the clean signal, σ is the noise standard deviation, and $z_i \sim \mathcal{N}(0, 1)$ is additive white Gaussian noise (AWGN).

Since DWT is a linear orthogonal transform, the noise remains white and spread evenly across all coefficients, whereas the signal energy is concentrated into a few large wavelet coefficients. Denoising is achieved by thresholding the DWT coefficients.

Hard and Soft Thresholding Formulations

Let w represent a wavelet coefficient, and λ be the threshold limit ($\lambda > 0$).

1. **Hard Thresholding:** Sets coefficients below the threshold to zero, keeping the others unchanged:

$$\eta_{\lambda}^{\text{hard}}(w) = \begin{cases} w, & |w| > \lambda \\ 0, & |w| \leq \lambda \end{cases}$$

2. **Soft Thresholding (Wavelet Shrinkage):** Sets coefficients below the threshold to zero, and shrinks the remaining coefficients towards zero:

$$\eta_{\lambda}^{\text{soft}}(w) = \text{sgn}(w) \max(|w| - \lambda, 0)$$

- **Comparison:** Hard thresholding preserves edge sharpness but can introduce spurious oscillations (pseudo-Gibbs phenomena) due to the discontinuity in the thresholding function. Soft thresholding is continuous, yielding smoother denoised signals, but it introduces a bias by scaling down large coefficients, which slightly blurs sharp peaks.

2.2 Threshold Selection Rules

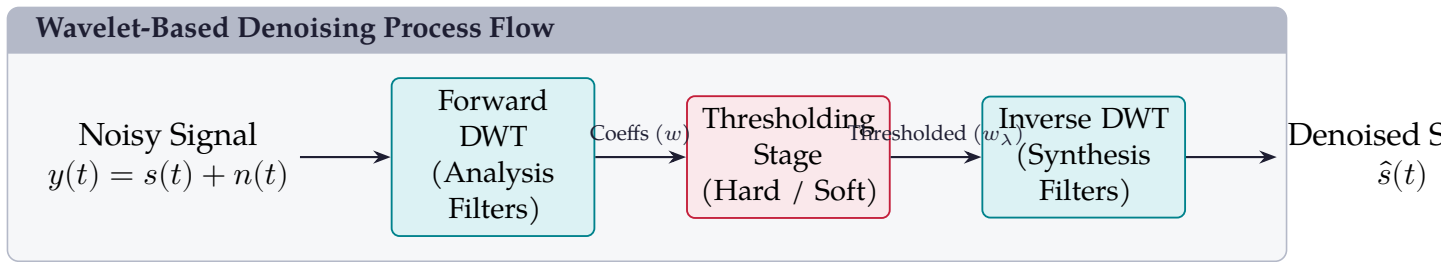
- **VisuShrink (Universal Threshold):** Uses a global threshold derived by Donoho and Johnstone:

$$\lambda_{\text{univ}} = \sigma \sqrt{2 \ln N}$$

where N is the length of the signal. It guarantees with high probability that the maximum noise coefficient is below λ . However, for large N , λ_{univ} becomes large, causing over-smoothing (loss of fine details). The noise variance σ is estimated from the median absolute deviation (MAD) of the finest detail subband (HH_1):

$$\hat{\sigma} = \frac{\text{median}(|w_{j,k}|)}{0.6745}, \quad w_{j,k} \in \text{finest subband}$$

- **SureShrink:** An adaptive, subband-dependent thresholding method. For each subband, it selects a threshold λ that minimizes **Stein's Unbiased Risk Estimate (SURE)**, optimizing the mean squared error (MSE) estimate.



2.3 Speckle Removal (Homomorphic Wavelet Filtering)

Speckle is a high-frequency multiplicative noise common in coherent imaging systems like Synthetic Aperture Radar (SAR) and Ultrasound:

$$y = x \cdot n$$

where y is the observed image, x is the noise-free image, and n is the speckle noise. Standard denoising assumes additive noise. To resolve this, **Homomorphic Filtering** is applied:

1. Take the logarithm of the noisy image to convert multiplicative noise into additive noise:

$$\ln(y) = \ln(x) + \ln(n) \implies y' = x' + n'$$

2. Apply DWT to y' , perform thresholding on the detail subbands, and apply IDWT to get the denoised estimate $\hat{x}'$.
3. Exponentiate the result to return to the original scale:

$$\hat{x} = \exp(\hat{x}')$$

2.4 Edge Detection and Object Isolation

- **Edge Detection:** Sharp changes in image intensity (edges) correspond to local maxima of the wavelet transform modulus at fine scales. By observing coefficients across multiple scales, we can distinguish true edges (which persist across scales) from random noise (which rapidly decays at coarser scales).
- **Image Fusion:** Combines complementary information from multiple images of the same scene (e.g., multi-focus or infrared/visible cameras). The source images are decomposed using DWT. The coefficient subbands are fused:
 - **LL (Approximation) Coefficients:** Typically averaged to preserve average illumination.
 - **Detail Coefficients (LH, HL, HH):** Selected using a "maximum absolute value" rule, as larger coefficients correspond to sharper, in-focus features.

Applying the IDWT to the fused coefficients yields a single high-quality composite image.

3. Discrete Wavelet Multi-Tone Modulation (DWMT)

Definition of DWMT

Discrete Wavelet Multi-Tone (DWMT) modulation is a multicarrier communication technique that uses Orthogonal Wavelet Filter Banks (specifically, Cosine-Modulated Filter Banks or Synthesis/Analysis Wavelet filters) as subcarrier generators instead of the DFT/IDFT used in standard OFDM.

In standard **OFDM (Orthogonal Frequency Division Multiplexing)**, subcarriers are shaped like sinc functions because they are modulated by a rectangular window. This yields high spectral sidelobes (-13 dB). Any frequency offset causes **Inter-Carrier Interference (ICI)**, requiring a **Cyclic Prefix (CP)** guard interval which reduces spectral efficiency by 10–25%.

3.0.1 Advantages of DWMT over OFDM:

1. **Low Sidelobes:** Wavelet filters have excellent spectral containment, with sidelobes decaying much faster than -13 dB (often below -40 dB).
2. **No Cyclic Prefix:** Because spectral overlap between non-adjacent subcarriers is negligible, DWMT does not require a cyclic prefix. This saves transmission bandwidth.
3. **High ICI/ISI Immunity:** The high stop-band attenuation of wavelet filters makes DWMT robust against narrowband interference and multipath channel distortions.

4. Beyond Wavelets: Ridgelets and Curvelets

4.1 Limitations of Wavelets in Two Dimensions

Standard 2D wavelets are created by tensor products of 1D wavelets (separable DWT). Their spatial support is square (isotropic). While excellent at representing point singularities, they struggle to capture **line or curve singularities** (such as edges in an image).

To represent a smooth curve, 2D wavelets require a large number of coefficients at fine scales, failing to achieve optimal **sparsity**. This limitation is known as the *curse of dimensionality* for directional features.

4.2 Ridgelet Transform

The continuous Ridgelet transform is designed specifically to capture line singularities in 2D. A ridgelet function in $\mathbb{R}^2$ is defined by scaling (a), translating (b), and rotating (θ):

$$\psi_{a,b,\theta}(x,y) = a^{-1/2} \psi\left(\frac{x \cos \theta + y \sin \theta - b}{a}\right)$$

Here, the wavelets are constant along the lines $x \cos \theta + y \sin \theta = \text{constant}$ (ridges).

Relationship to Radon Transform: The Ridgelet transform can be implemented by first taking the **Radon Transform** of the image $f(x,y)$ (which maps line integrals to points) and then applying a **1D Wavelet Transform** along the radial direction:

$$\text{Ridgelet Transform} = \text{1D DWT} \circ \text{Radon Transform}$$

4.3 Curvelet Transform

While Ridgelets handle straight line singularities, real-world images contain curved edges. The **Curvelet Transform** generalizes Ridgelets to handle curves by partitioning the image into small local patches, and applying a local Ridgelet transform to each patch.

Parabolic Scaling of Curvelets

A key mathematical property of curvelets is **Parabolic Scaling**, which governs the relationship between the width and length of their spatial support at scale j :

$$\text{width} \approx \text{length}^2 \quad \Rightarrow \quad \text{width} \propto 2^{-j}, \quad \text{length} \propto 2^{-j/2}$$

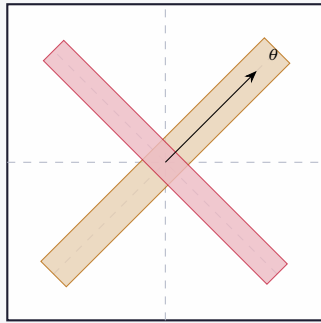
At fine scales, curvelets become highly needle-like (highly anisotropic), allowing them to align perfectly with curved edge structures.

4.3.1 Digital Curvelet Transform Implementation

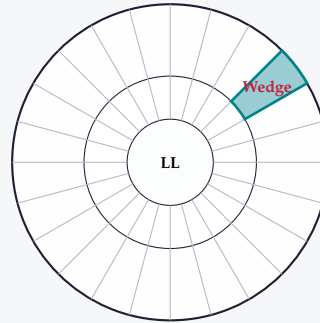
Two fast algorithms exist for the discrete curvelet transform:

1. **Unequally Spaced Fast Fourier Transform (USFFT)**: Samples the Fourier transform of the image along a polar-like grid of wedges.
2. **Wrapping Algorithm**: Wraps Fourier wedges into a rectangular grid around the origin, which is faster and computationally more efficient.

Ridgelet vs. Curvelet Frequency Domain Partitioning



(a) Ridgelet: Radial Slabs



(b) Curvelet: Polar Wedges

Feature	Wavelets	Ridgelets	Curvelets
Target Singularity	Point singularities.	Straight line singularities.	Smooth curved singularities.
Support Shape	Isotropic (square/circular).	Infinite line strip (ridges).	Needle-like curved segments.
Directional Sensitivity	Limited (horizontal, vertical, diagonal).	Continuous directional angle θ .	Multi-directional (increases at fine scales).
Scaling Law	$width \approx length$	$width \propto scale$	$width \approx length^2$ (Parabolic).
Sparsity for Curves	Poor (requires many coefficients).	Poor (struggles with curved features).	Optimal (highly sparse representation).

Quick Revision — Module 5

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Key Equations

Hard Thresholding	$\eta_{\lambda}^{\text{hard}}(w) = w \cdot \mathbb{I}(w > \lambda)$
Soft Thresholding	$\eta_{\lambda}^{\text{soft}}(w) = \text{sgn}(w) \max(w - \lambda, 0)$
Universal Threshold	$\lambda_{\text{univ}} = \sigma\sqrt{2 \ln N}$
Ridgelet Formula	$\psi_{a,b,\theta}(x, y) = a^{-1/2} \psi\left(\frac{x \cos \theta + y \sin \theta - b}{a}\right)$
Parabolic Scaling Law	$\text{width} \propto \text{length}^2 \implies w \propto 2^{-j}, l \propto 2^{-j/2}$

One-Line Definitions

- **Energy Compaction:** Concentrate signal representation into a small number of large coefficients.
- **Blocking Artifacts:** Grid-like boundaries visible at low bitrates in block-based DCT compression.
- **Wavelet Shrinkage:** Noise reduction method that attenuates coefficients toward zero (soft thresholding).
- **VisuShrink:** Denoising method using the universal threshold $\sigma\sqrt{2 \ln N}$ to remove noise with high probability.
- **SureShrink:** Adaptive denoising scheme minimizing Stein's Unbiased Risk Estimate for each subband.
- **Homomorphic Filter:** Transform mapping multiplicative noise to additive noise via log operations.
- **DWMT Modulation:** Multicarrier modulation using wavelet synthesis/analysis filter banks instead of DFT.
- **Ridgelet Transform:** Directional 2D transform mapping line singularities to point-like representations.
- **Curvelet Transform:** Directional multiscale transform optimized for curved edge singularities.

Frequently Asked / Viva Questions

Q1. Why does JPEG 2000 perform better than standard JPEG at low bitrates?

Standard JPEG uses block-based DCT (8×8), which causes blocking artifacts due to independent processing of boundaries. JPEG 2000 uses DWT globally on the entire image, which yields no block borders and results in a graceful blur at low bitrates.

Frequently Asked / Viva Questions (contd.)

Q2. Compare hard and soft thresholding in terms of visual quality and math.

Hard thresholding is discontinuous and keeps coefficients larger than λ unchanged, which preserves sharp edges but can create pseudo-Gibbs oscillations. Soft thresholding is continuous and reduces coefficients by λ , which yields smoother signals but introduces bias and blurs sharp peaks.

Q3. What is the universal threshold in VisuShrink, and what is its drawback?

$\lambda_{\text{univ}} = \sigma\sqrt{2 \ln N}$. Its drawback is that as signal length N increases, the threshold becomes excessively large. This removes almost all noise but also removes true high-frequency signal details, causing over-smoothing.

Q4. How do we estimate the noise variance σ in practical image denoising?

It is estimated from the median absolute deviation (MAD) of the finest detail subband coefficients (HH_1): $\hat{\sigma} = \text{median}(|w_{j,k}|)/0.6745$.

Q5. Explain the homomorphic wavelet denoising workflow for speckle noise.

Speckle is multiplicative ($y = x \cdot n$). We apply a logarithmic transform: $\ln(y) = \ln(x) + \ln(n)$ to convert it to additive noise. We then apply DWT, perform wavelet thresholding, apply IDWT, and finally exponentiate the result.

Q6. How is DWT used for image fusion?

Multiple source images are decomposed by DWT. The LL (approximation) coefficients are averaged to preserve illumination, and the detail coefficients (LH, HL, HH) are fused by taking the maximum absolute value (which preserves sharp details and focus). The fused image is reconstructed using IDWT.

Q7. What are the advantages of DWMT over OFDM?

DWMT uses wavelet filter banks which have much lower spectral sidelobes than OFDM's sinc subcarriers. This reduces Inter-Carrier Interference (ICI) and eliminates the need for a Cyclic Prefix (CP), increasing spectral efficiency.

Q8. Why are standard wavelets inefficient at representing curved edges in 2D?

Separable 2D wavelets have square spatial support. Representing a smooth curved edge requires many small wavelets along the curve at fine scales, which destroys sparsity.

Q9. How is the Ridgelet transform implemented using the Radon transform?

The Ridgelet transform is implemented by first applying the Radon transform to project line integrations into 1D profiles, followed by a 1D continuous wavelet transform along those radial profiles.

Q10. What is the parabolic scaling property of curvelets?

Curvelets scale such that width $\approx$ length² (i.e. at scale 2^{-j} , the width is 2^{-j} and length is $2^{-j/2}$). This anisotropic, needle-like shape allows curvelets to trace curved edges efficiently.

Q11. Explain the two primary digital curvelet transform algorithms.

Frequently Asked / Viva Questions (contd.)

- **USFFT:** Computes spatial coordinates on an irregular polar-like grid in the frequency domain.
- **Wrapping:** Wraps rectangular frequency wedges into a standard rectangular grid to allow fast FFT processing.

Q12. What are the primary applications of Ridgelets and Curvelets?

They are used in seismic data processing, CT/MRI tomographic reconstruction, and denoising of images with dominant curved structures, where wavelets would blur the edges.

Summary Matrix: Multi-Resolution Feature Representation

Property	Wavelet Transform	Ridgelet Transform	Curvelet Transform
Dimension	1D, 2D, or Multi-D.	2D (or higher).	2D (or higher).
Optimal For	Point singularities.	Straight line singularities.	Curved singularities (edges).
Support Geometry	Isotropic (square).	Line-like (infinite slabs).	Needle-like (parabolic wedges).
Directionality	3 fixed directions in 2D.	Continuous angle θ .	Multi-directional (scale-dependent).
Fourier Tiling	Concentric dyadic squares.	Radial strips.	Polar-like wedges.